

Step 1: Create a work directory for building and compiling your SELinux modules.

Instead of working in default directories, experience has taught me to work as a non-root user in non-default folders. Let us log in as the non-root user and make a working directory for our SELinux modules:

```
[vbg@vbg ~]$ mkdir test-selinux
[vbg@vbg ~]$ cd test-selinux/
```

Step 2: Copy the necessary files needed for compiling SELinux modules.

The only important file that you need to copy to your working directory is the Makefile from the `/usr/share/selinux/devel/` directory.

```
[vbg@vbg test-selinux]$ cp /usr/share/selinux/devel/Makefile .
[vbg@vbg test-selinux]$ ls
```

Step 3: Create at least a .te file for your SELinux module, specifying the module name and version number:

```
[vbg@vbg test-selinux]$ vim test.te
```

Put the following text in the above file:

```
policy_module(test,1.0)

type ify_t;
```

And save the source file.

The above step specifies that we are creating a policy module named `test` with the version 1.0 and are declaring a new type to be introduced in the policy called `ify_t`.

Step 4: Compile the source file above to a binary policy package file

To compile the source, simply run `make` (ensure you have copied the Makefile in Step 2 shown above):

```
[vbg@vbg test-selinux]$ make
Compiling targeted test module
/usr/bin/checkmodule: loading policy
```

```
configuration from tmp/test.mod
/usr/bin/checkmodule: policy configuration
loaded
/usr/bin/checkmodule: writing binary
representation (version 6) to tmp/test.mod
Creating targeted test.pp policy package
rm tmp/test.mod.fc tmp/test.mod
```

You should now have the compiled policy package file `test.pp`:

```
[vbg@vbg test-selinux]$ ll
total 60
-rw-r--r-- 1 vbg vbg 437 Sep 24 10:36 Makefile
-rw-rw-r-- 1 vbg vbg 0 Sep 24 10:36 test.fc
-rw-rw-r-- 1 vbg vbg 0 Sep 24 10:36 test.if
-rw-rw-r-- 1 vbg vbg 22994 Sep 24 10:36 test.pp
-rw-rw-r-- 1 vbg vbg 37 Sep 24 10:36 test.te
drwxrwxr-x 2 vbg vbg 4096 Sep 24 10:36 tmp
```

To check if a type `ify_t` exists, just use the following command:

```
[root@vbg devel]# seinfo -l | grep ify
```

You will not receive any output. This shows that the type `ify_t` does not exist in our policy currently.

Step 5: Load the binary policy package.

As the root user, use the `semodule` command to load the compiled `test.pp` policy package. The option to use with the `semodule` command is `-i` (which stands for insert).

```
[root@vbg devel]# semodule -i /home/vbg/test-selinux/test.pp
```

To confirm if the module has been successfully loaded, list all the modules:

```
[root@vbg devel]# semodule -l
ccs 1.0.0
clamav 1.1.0
doc 1.1.0
evolution 1.1.0
```

```
iscsid 1.0.0
mozilla 1.1.0
mplayer 1.1.0
nagios 1.1.0
odddjob 1.0.1
openoff 1.0.0
pcscd 1.0.0
pyzor 1.1.0
razor 1.1.0
ricci 1.0.0
smartmon 1.1.0
test 1.0
tmp 1.0.1
vbg 1.0.3
```

You can see that a module named `test` with version 1.0 has been loaded. To check whether our new type has been added, retype the earlier command:

```
[root@vbg devel]# seinfo -l | grep ify
ify_t
```

This shows that we have successfully introduced a new type in our SELinux policy by adding a policy module.

Up next

In this article, we have covered the basics of policy modules. In the next part of the series, we will look at adding allow rules, type transition rules, Booleans and other policy building blocks. We shall also look at the `audit2allow` tool that helps to create policy modules.

I hope the article has helped you understand the modular structure of SELinux. It is this modularity that allows administrators to easily create SELinux policy modules for whatever applications they deploy -- without compromising the security of their system.

Looking forward to meeting you all again, next month... 

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